PassFrame tuition

Fraction Operations

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: 1/2 + 1/4

Method & answer: Rewrite 1/2 as 2/4 (since 4 is a multiple of 2). Then 2/4 + 1/4 = 3/4.

Standard: 1/3 + 1/6

Method & answer: 6 is a multiple of 3, so rewrite 1/3 as 2/6. Then 2/6 + 1/6 = 3/6 (= 1/2).

Advanced: Multiply 1/2 by 1/3.

Method & answer: Multiply the numerators (1×1=1) and the denominators (2×3=6): 1/2 × 1/3 = 1/6.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. 1/2 + 1/4

2. Which fraction is equivalent to 1/3 with a denominator of 6? A) 1/6 B) 2/6 C) 3/6 D) 6/3

3. 2/3 − 1/6

4. Multiply 1/4 by 1/2.

5. What is 1/2 + 1/8? A) 2/8 B) 5/8 C) 1/16 D) 3/8

6. 3/4 − 1/8

7. A recipe needs 1/2 cup of flour and 1/4 cup of sugar. What is the total amount of flour and sugar needed?

8. Select ALL the pairs below that can be added by first changing only ONE fraction: A) 1/2 and 1/4 B) 1/3 and 1/5 C) 1/5 and 1/10 D) 1/4 and 1/5

9. 1/3 + 1/9

10. Multiply 1/5 by 1/2. A) 1/10 B) 2/5 C) 1/7 D) 2/10

11. 5/6 − 1/3

12. 1/4 + 3/8

13. A jug has 1/2 litre of water and another jug has 3/8 litre. How much water is there altogether?

14. What is 7/10 − 1/5? A) 5/10 B) 6/10 C) 1/10 D) 8/10

15. 2/3 + 1/12

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. 1/4 + 1/6

2. Work out 2/5 + 1/4. A) 3/9 B) 13/20 C) 3/20 D) 6/20

3. 5/6 − 1/4

4. Multiply 2/3 by 1/4.

6. 2/3 − 1/5

7. A tank is filled with 2/5 oil and 1/3 water, by volume. What fraction of the tank is filled in total?

8. Select ALL the calculations below that have the same answer as 1/2 + 1/3: A) 3/6 + 2/6 B) 5/6 C) 1/6 × 5 D) 4/6 + 2/6

9. 3/8 + 1/4

11. 7/8 − 1/2

12. 5/9 − 1/3

13. A gardener plants vegetables on 3/5 of a plot and flowers on 1/4 of the plot. What fraction of the plot is planted in total?

14. Which of these equals 1/3 × 3/4,

in simplest form? A) 4/12 B) 3/12 C) 1/4 D) 4/3

5. What is 3/4 × 2/5, in its simplest form? A) 6/20 B) 3/10 C) 5/9 D) 5/20

10. Multiply 3/5 by 1/2. A) 3/10 B) 4/10 C) 3/7 D) 2/10

15. 2/5 + 3/10 − 1/10

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A tank is 1/3 full of water. Another 1/4 tank of water is poured in. What fraction of the tank is now full?

6. Which of these is equal to 5/6 − 2/9? A) 3/18 B) 11/18 C) 7/18 D) 1/3

11. Convert 11/4 to a mixed number, then subtract 1/4.

2. What is 5/6 + 3/4, written as a mixed number? A) 19/12 B) 1 7/12 C) 1 5/10 D) 8/10

7. A charity raised 5/6 of its target in week one and a further 1/12 of its target in week two. What fraction of the target has it raised in total?

12. A water butt is 2/3 full. A hosepipe adds 1/6 more, then 1/4 is used for watering plants. What fraction of the butt is full at the end?

3. 2/5 + ? = 9/10. What fraction is missing?

8. Select ALL of the following that are equivalent to 3/4 × 1/3: A) 3/12 B) 1/4 C) 1/3 D) 6/24

13. Which TWO of these calculations give an answer greater than 1 whole? A) 3/4 + 5/6 B) 7/8 + 1/3 C) 1/2 + 1/3 D) 2/5 + 1/4

4. Multiply 3/4 by 2/3, then simplify.

9. 7/10 − 1/2, then add 1/5 to your answer.

14. A tailor needs 5/6 m of cloth for a shirt and uses 2/3 of that amount for a smaller version. How much cloth does the smaller version need?

5. A recipe uses 3/4 cup of flour for one batch. How much flour is needed for 2/3 of a batch?

10. What is 2/3 × 3/5, in simplest form? A) 6/15 B) 2/5 C) 5/8 D) 6/8

15. 9/10 − 3/5, then multiply your answer by 1/2.